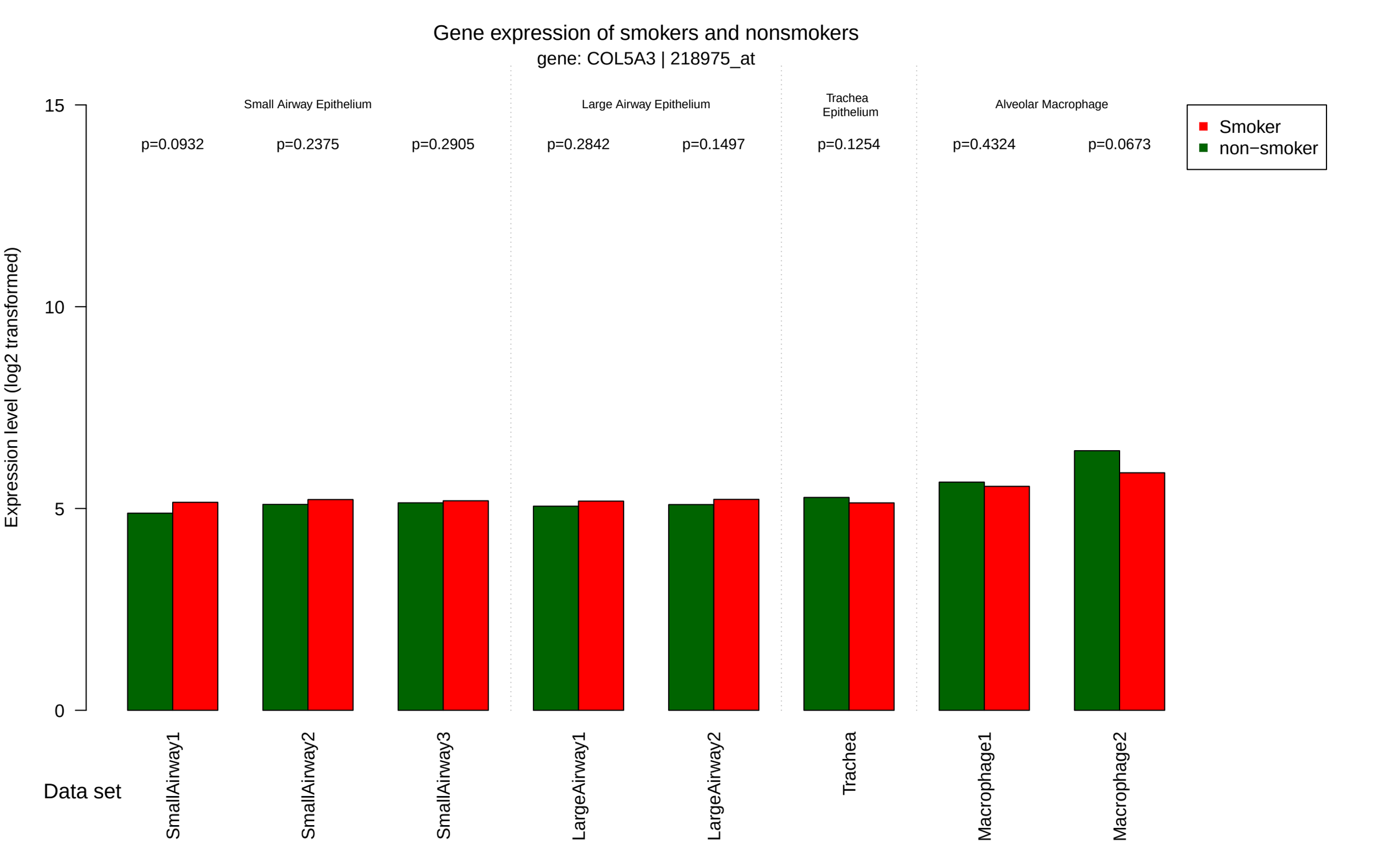
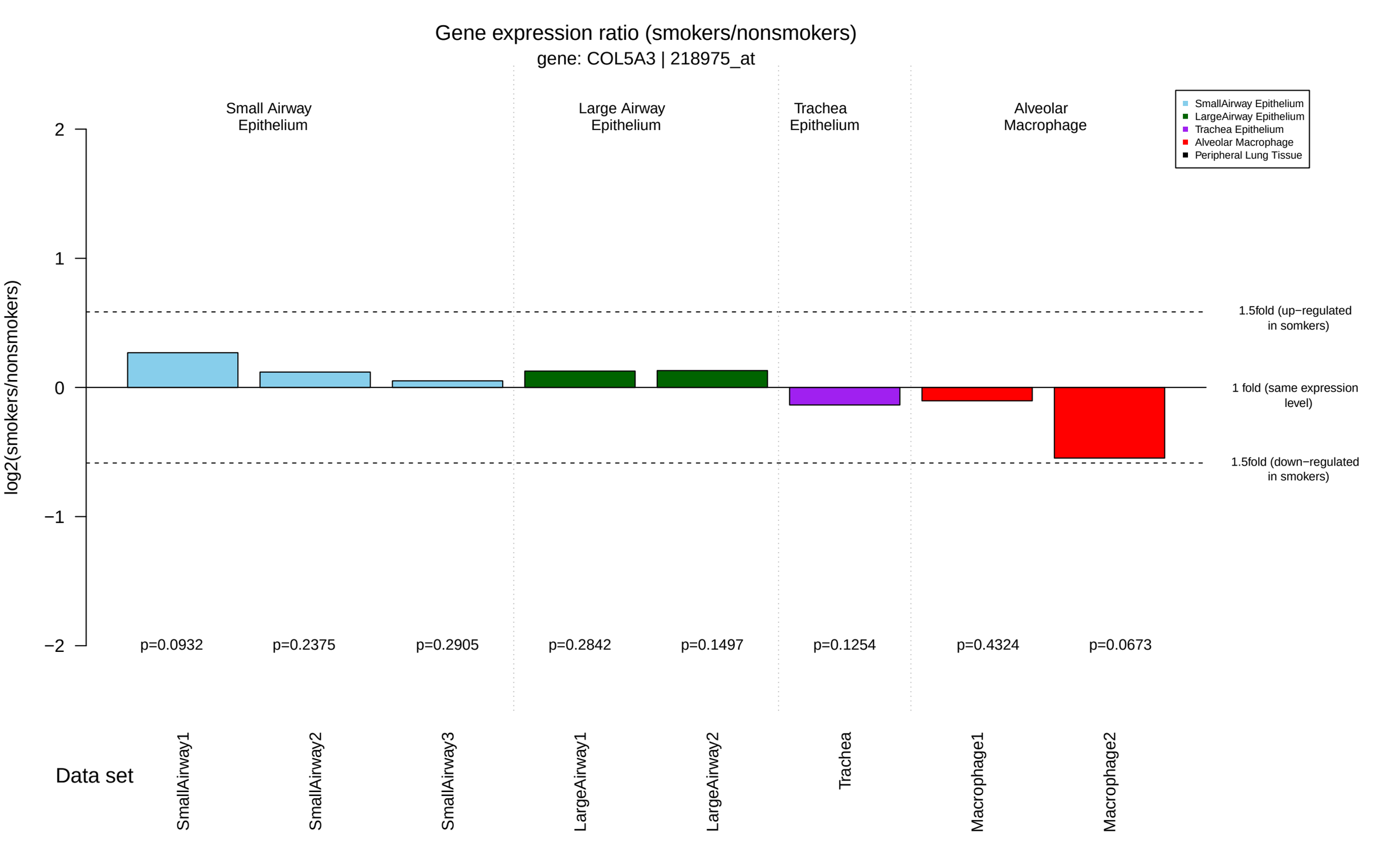
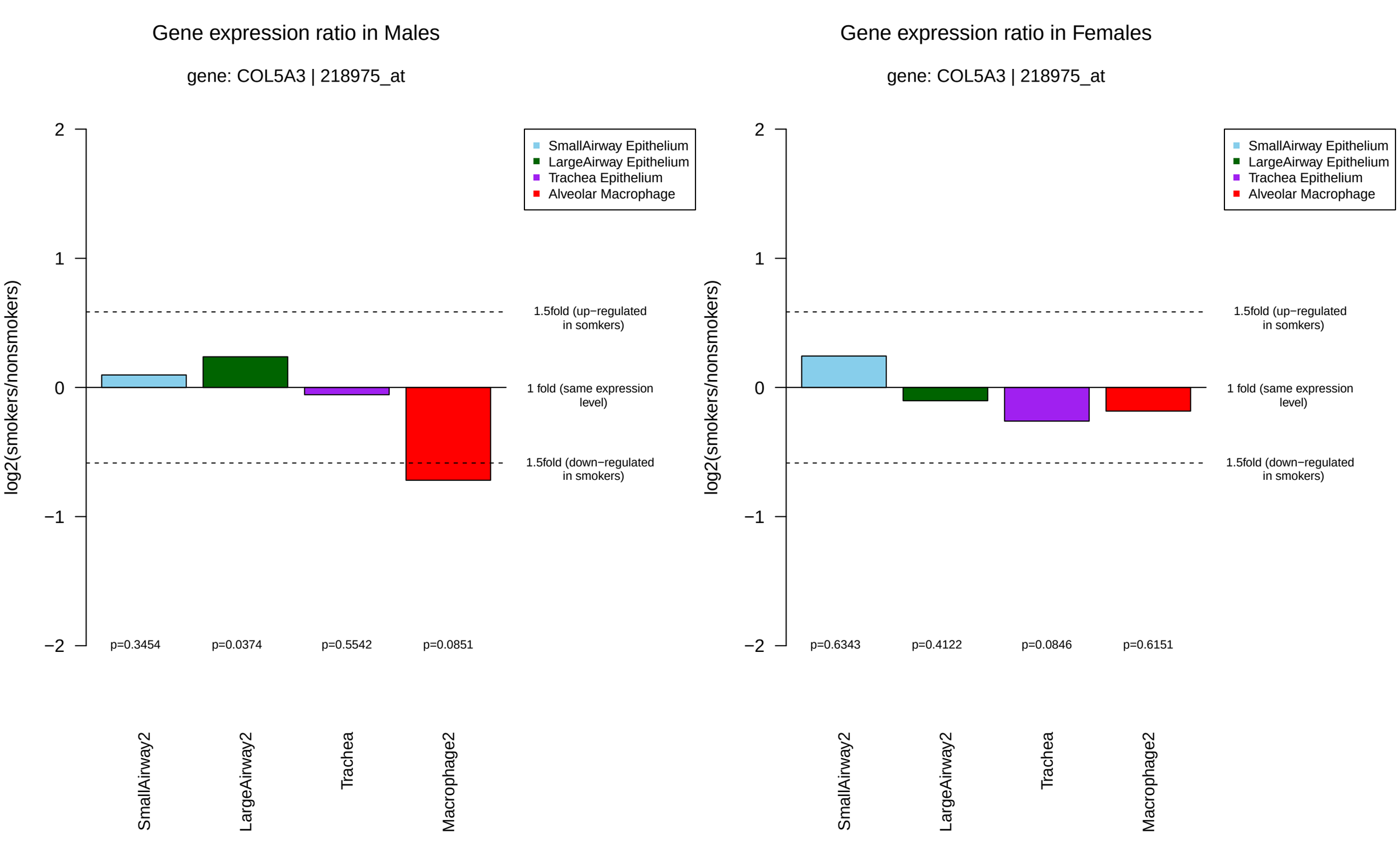
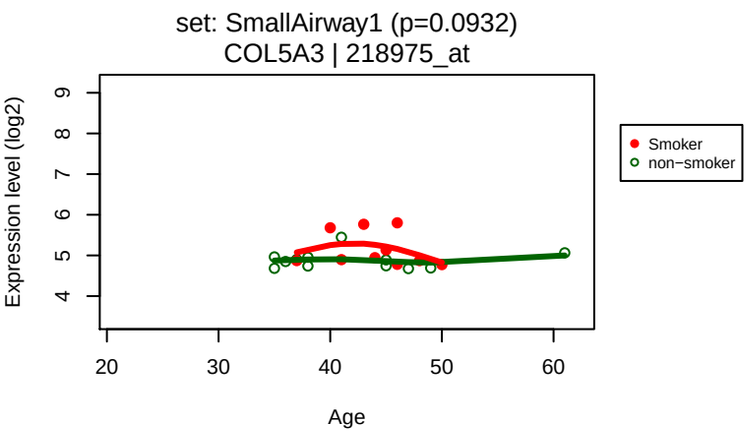
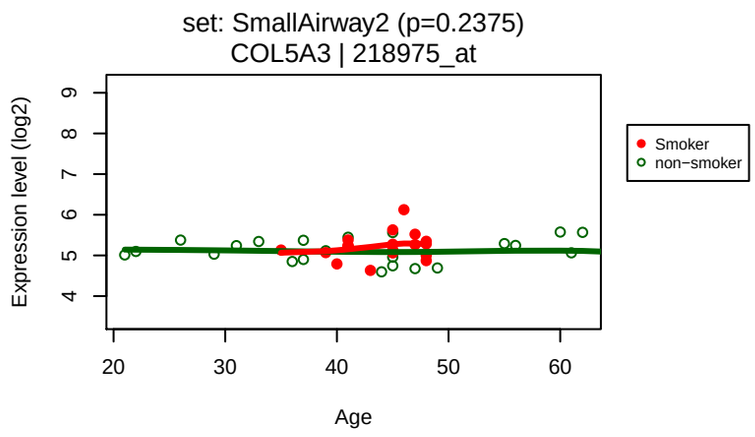
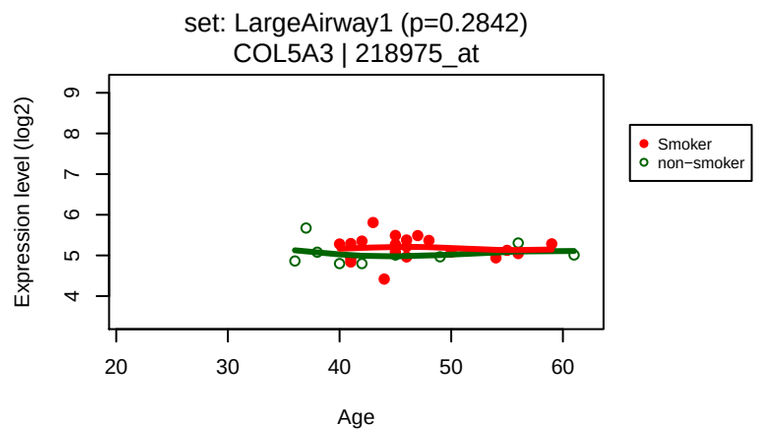

Gene expression of smokers and nonsmokers
gene: COL5A3 | 218975_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0932
p=0.2375
p=0.2905
p=0.2842
p=0.1497
p=0.1254
p=0.4324
p=0.0673
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

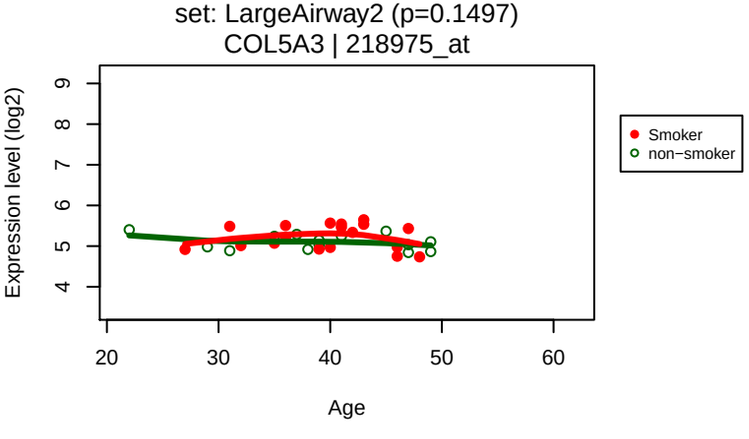

Gene expression ratio (smokers/nonsmokers)
gene: COL5A3 | 218975_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0932
p=0.2375
p=0.2905
p=0.2842
p=0.1497
p=0.1254
p=0.4324
p=0.0673
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

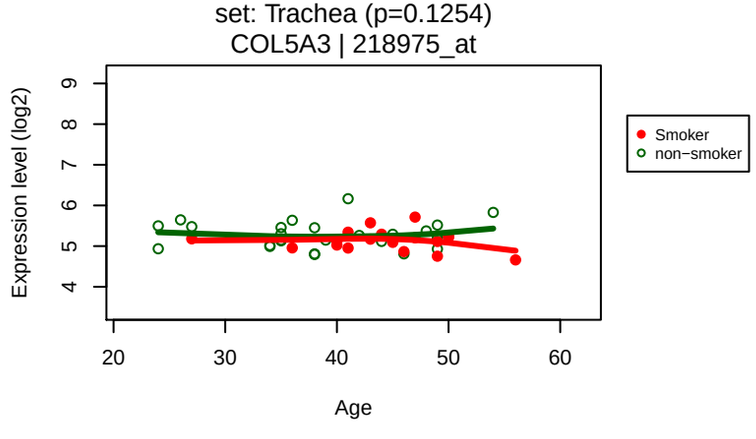

Gene expression ratio in Males
gene: COL5A3 | 218975_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3454
p=0.0374
p=0.5542
p=0.0851
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL5A3 | 218975_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6343
p=0.4122
p=0.0846
p=0.6151
SmallAirway2
LargeAirway2
Trachea
Macrophage2

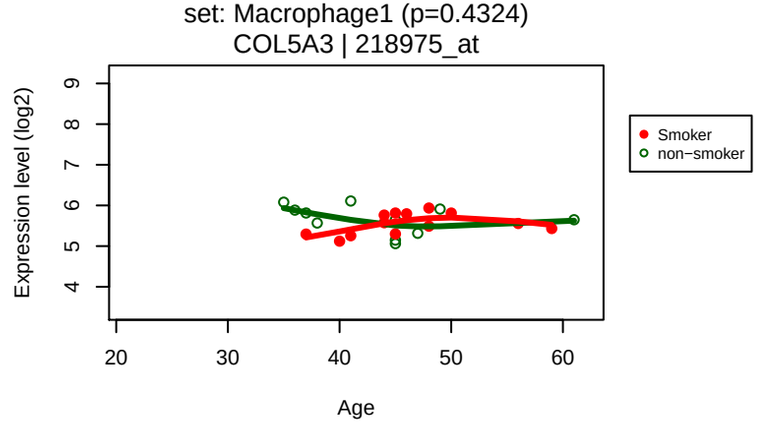

set: SmallAirway1 (p=0.0932)
COL5A3 | 218975_at
Expression level (log2)
Age
Smoker
non−smoker

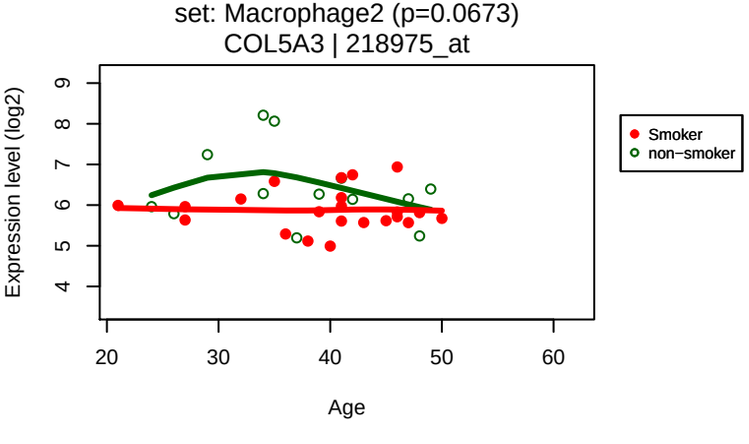

set: SmallAirway2 (p=0.2375)
COL5A3 | 218975_at
Expression level (log2)
Age
Smoker
non−smoker

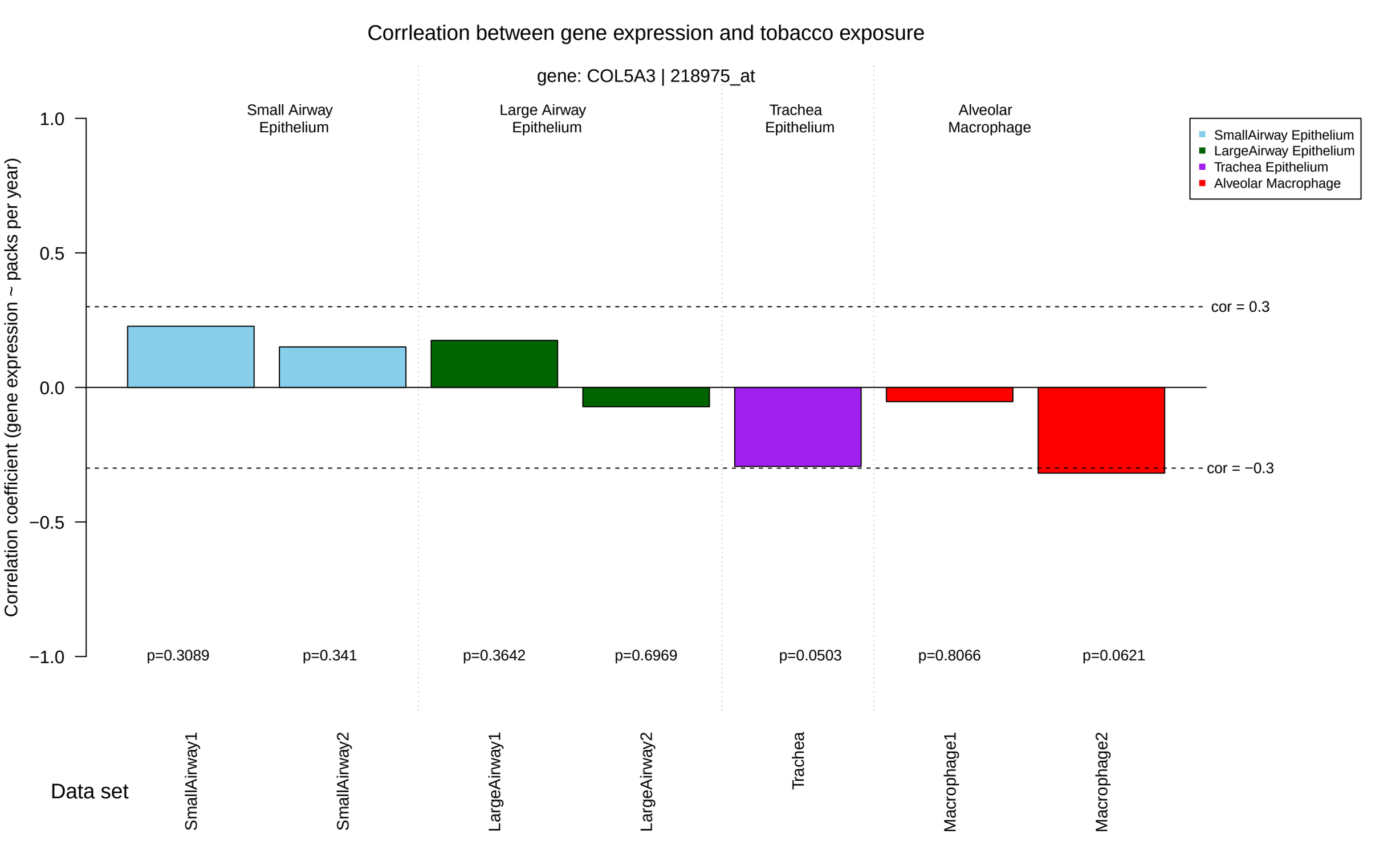

set: LargeAirway1 (p=0.2842)
COL5A3 | 218975_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1497)
COL5A3 | 218975_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1254)
COL5A3 | 218975_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4324)
COL5A3 | 218975_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0673)
COL5A3 | 218975_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL5A3 | 218975_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3089
p=0.341
p=0.3642
p=0.6969
p=0.0503
p=0.8066
p=0.0621
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2